

Safety data sheet

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BASF Safety data sheet
Date / Revised: 21.09.2022
Product: **Trilon® M Liquid**

Version: 5.0

(30043459/SDS_GEN_TH/EN)

Date of print): 08.07.2024

1. Substance/preparation and manufacturer/supplier identification

Product name:
Trilon® M Liquid

Use: complexing agents for the chemical industry

Manufacturer/supplier:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Care Chemicals
Telephone: +49 621 60-57579
E-mail address: emd-ems-ehs-masterdata@basf.com

Emergency information:
International emergency number:
Telephone: +49 180 2273-112

2. Hazard identification

Classification according to UN GHS 2009

Classification of the substance and mixture:
| No need for classification according to GHS criteria for this product.

Label elements and precautionary statement:

| The product does not require a hazard warning label in accordance with GHS criteria.

Other hazards which do not result in classification:
No specific dangers known, if the regulations/notes for storage and handling are considered.

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3. Composition/information on ingredients

Chemical nature

Substance nature: mixture

Alanine, N,N-bis(carboxymethyl)-, trisodium salt
CAS - Nr: 164462-16-2

in water

Hazardous ingredients

sodium hydroxide

Content (W/W): $\geq 0.3\%$ - $< 1\%$
CAS Number: 1310-73-2

Met. Corr.: Cat. 1
Skin Corr./Irrit.: Cat. 1A
Eye Dam./Irrit.: Cat. 1

4. First-Aid Measures

General advice:

Remove contaminated clothing.

If inhaled:

Keep patient calm, remove to fresh air.

On skin contact:

Wash thoroughly with soap and water

On contact with eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:

Rinse mouth and then drink 200-300 ml of water.

Note to physician:

Symptoms: (Further) symptoms and / or effects are not known so far

Treatment: Symptomatic treatment (decontamination, vital functions).

5. Fire-Fighting Measures

Suitable extinguishing media:

water spray, dry powder, foam

Specific hazards:

harmful vapours, nitrous gases, carbon oxides

Evolution of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

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Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
Contaminated extinguishing water must be disposed of in accordance with official regulations.

6. Accidental Release Measures

Personal precautions:
Use personal protective clothing. Information regarding personal protective measures, see section 8.

Environmental precautions:
Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:
For large amounts: Dike spillage. Pump off product.
For residues: Pick up with suitable absorbent material.
Dispose of absorbed material in accordance with regulations.

Additional information: High risk of slipping due to leakage/spillage of product.

7. Handling and Storage

Handling

No special measures necessary provided product is used correctly.

Protection against fire and explosion:
No special precautions necessary.

Storage

Suitable materials for containers: High density polyethylene (HDPE), Low density polyethylene (LDPE), Stainless steel 1.4301 (V2), Stainless steel 1.4305, Stainless steel 1.4306 (V2A), Stainless steel 1.4361, Stainless steel 1.4401, Stainless steel 1.4404, Stainless steel 1.4439, Stainless steel 1.4539, Stainless steel 1.4541, Stainless steel 1.4571

Further information on storage conditions: Keep container tightly closed and in a cool place.

Storage stability:
Storage temperature: 0 - 50 °C

Protect from temperatures below: -20 °C
The packed product is not damaged by low temperatures or by frost.
Protect from temperatures above: 50 °C
The packed product will not be damaged by high temperatures.

8. Exposure controls and personal protection

Components with occupational exposure limits

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sodium hydroxide, 1310-73-2;
CLV 2 mg/m³ (ACGIHTLV)
TWA value 2 mg/m³ (OEL (TH))

Personal protective equipment

Respiratory protection:

Respiratory protection in case of vapour/aerosol release. Particle filter with medium efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P2 or FFP2)

Hand protection:

Chemical resistant protective gloves

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):

e.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), polyvinylchloride (0.7 mm) and other

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:

Safety glasses with side-shields.

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:

Wearing of closed work clothing is recommended. No eating, drinking, smoking or tobacco use at the place of work. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

Form:	liquid
Colour:	yellowish
Odour:	product specific
Odour threshold:	not determined

pH value:	approx. 11.3 (1 %(m), 25 °C)	(DIN 19268)
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solidification point:	-50 °C	(DIN ISO 3016)
Boiling point:	approx. > 100 °C (1,013 hPa)	

Information on: water	
Boiling point:	100 °C

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Flash point:

A flash point determination is unnecessary due to the high water content.

Evaporation rate:

not determined

Flammability (solid/gas): not self-igniting

Lower explosion limit:

For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.

Upper explosion limit:

For liquids not relevant for classification and labelling.

Ignition temperature:

> 200 °C

Thermal decomposition: > 200 °C

(DTA)

Self ignition:

not self-igniting

Self heating ability:

It is not a substance capable of spontaneous heating according to UN transport regulations class 4.2.

Explosion hazard: not explosive

Fire promoting properties: not fire-propagating

Vapour pressure:

< 0.1 hPa
(20 °C)

Information on: water

Vapour pressure:

23.4 hPa
(20 °C)
Literature data.

Density:

1.30 g/cm³
(20 °C)
1.28 g/cm³
(50 °C)

Relative density:

1.27 - 1.33
(20 °C)

Relative vapour density (air):

not determined

Solubility in water:

soluble
(20 °C)

Partitioning coefficient n-octanol/water (log Pow):

Study does not need to be conducted.

Surface tension:

No data available.

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Viscosity, dynamic:	approx. 25 mPa.s (20 °C)	(DIN EN 12092)
	< 20 mPa.s (50 °C)	(DIN EN 12092)

Other Information:

If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity

Conditions to avoid:

See SDS section 7 - Handling and storage.

Thermal decomposition: > 200 °C (DTA)

Substances to avoid:

amphoteric metals, light metals

Corrosion to metals:

Corrosive effect on:
Aluminium

Based on available data, the classification criteria are not met.

Corrosion to metals:

Not corrosive to:
steel

Hazardous reactions:

No hazardous reactions when stored and handled according to instructions.

Hazardous decomposition products:

No hazardous decomposition products if stored and handled as prescribed/indicated.

Chemical stability:

The product is stable if stored and handled as prescribed/indicated.

11. Toxicological Information

Routes of exposure

Acute oral toxicity

Experimental/calculated data:

LD50rat (oral): > 2,000 mg/kg (Guideline 92/69/EEC, B.1)

Acute inhalation toxicity

LC50 rat (by inhalation):

not determined

Acute dermal toxicity

LD50 rat (dermal): > 2,000 mg/kg (OECD Guideline 402)

Assessment of acute toxicity

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Virtually nontoxic after a single ingestion. Virtually nontoxic after a single skin contact.

Information on: Alanine, N,N-bis(carboxymethyl)-, trisodium salt

Acute oral toxicity

Experimental/calculated data:

LD50 rat (oral): > 2,000 mg/kg (Guideline 92/69/EEC, B.1)

Limit concentration test only (LIMIT test). No mortality was observed.

Information on: Alanine, N,N-bis(carboxymethyl)-, trisodium salt

Acute dermal toxicity

Experimental/calculated data:

LD50 rat (dermal): > 2,000 mg/kg (OECD Guideline 402)

Limit concentration test only (LIMIT test). No mortality was observed.

Symptoms

(Further) symptoms and / or effects are not known so far

Irritation

Assessment of irritating effects:

Not irritating to eyes and skin.

Information on: Alanine, N,N-bis(carboxymethyl)-, trisodium salt

Experimental/calculated data:

Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404)

Information on: Alanine, N,N-bis(carboxymethyl)-, trisodium salt

Experimental/calculated data:

Serious eye damage/irritation rabbit: non-irritant (OECD Guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:

Skin sensitizing effects were not observed in animal studies.

Information on: Alanine, N,N-bis(carboxymethyl)-, trisodium salt

Experimental/calculated data:

Guinea pig maximization test guinea pig: Non-sensitizing. (OECD Guideline 406)

Germ cell mutagenicity

Assessment of mutagenicity:

Based on available data, the classification criteria are not met.

Experimental/calculated data:

Ames-test

negative (OECD Guideline 471)

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Ames-test
negative (OECD Guideline 472)

Cytogenetic assay
The genotoxic effect shown in vitro was not confirmed in vivo. (OECD Guideline 473)

HGPRT assay
negative (OECD Guideline 476)

Micronucleus assay
negative (OECD Guideline 474)

Information on: Alanine, N,N-bis(carboxymethyl)-, trisodium salt

Assessment of mutagenicity:

The substance was not mutagenic in bacteria. The substance induced chromosomal aberrations in a mammalian cell culture test. The substance was not mutagenic in a test with mammals.

Carcinogenicity

Assessment of carcinogenicity:

Based on available data, the classification criteria are not met.

In long-term studies in rats in which the substance was given by feed, a carcinogenic effect was not observed. The data on toxicology refer to the active ingredient.

Experimental/calculated data:

rat (oral feed)

Result: negative

Reproductive toxicity

Assessment of reproduction toxicity:

Based on available data, the classification criteria are not met.

Information on: Alanine, N,N-bis(carboxymethyl)-, trisodium salt

Assessment of reproduction toxicity:

The results of animal studies gave no indication of a fertility impairing effect. The results were determined in a Screening test (OECD 421/422).

Developmental toxicity

Assessment of teratogenicity:

Based on available data, the classification criteria are not met.

Information on: Alanine, N,N-bis(carboxymethyl)-, trisodium salt

Assessment of teratogenicity:

No indications of a developmental toxic / teratogenic effect were seen in animal studies.

Specific target organ toxicity (single exposure)

Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

The substance may cause damage to the kidney after repeated ingestion of high doses, as shown in animal studies. The data on toxicology refer to the active ingredient.

Based on available data, the classification criteria are not met.

Aspiration hazard

No aspiration hazard expected.

12. Ecological Information

Ecotoxicity

Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish:

LC50 (96 h) > 200 mg/l, *Brachydanio rerio* (OECD 203; ISO 7346; 92/69/EEC, C.1)

Aquatic invertebrates:

EC50 (48 h) > 200 mg/l, *Daphnia magna* (OECD Guideline 202, part 1)

Aquatic plants:

EC50 (72 h) > 200 mg/l (biomass), *Scenedesmus subspicatus* (Guideline 92/69/EEC, C.3, static)

Limited influence on algae growth due to chelate formation.

Microorganisms/Effect on activated sludge:

EC20 (0.5 h) > 2,000 mg/l, activated sludge, domestic (OECD Guideline 209, aerobic)

Chronic toxicity to fish:

No observed effect concentration (28 d) > = 200 mg/l, *Oncorhynchus mykiss* (OECD Guideline 204)

Chronic toxicity to aquatic invertebrates:

No observed effect concentration (21 d), >= 200 mg/l, *Daphnia magna* (OECD Guideline 202, part 2, semistatic)

Assessment of terrestrial toxicity:

No data available concerning terrestrial toxicity.

Soil living organisms:

LC50 (14 d) 300 mg/kg, *Eisenia foetida* (OECD Guideline 207, artificial soil)

Terrestrial plants:

EC50 (19 d) 1.600 mg/kg, *Avena sativa* (OECD Guideline 208)

Mobility

Assessment transport between environmental compartments:
The substance will not evaporate into the atmosphere from the water surface.
Adsorption to solid soil phase is not expected.

Persistence and degradability

Assessment biodegradation and elimination (H₂O):
Readily biodegradable (according to OECD criteria).

Elimination information:
> 60 % BOD of the ThOD (28 d) (OECD Guideline 301 F) (aerobic)

Based on OECD criteria the product is readily biodegradable.

Bioaccumulation potential

Assessment bioaccumulation potential:
The product has not been tested.

Bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

Additional information

Add. remarks environm. fate & pathway:
Treatment in biological waste water treatment plants has to be performed according to local and administrative regulations.

Other ecotoxicological advice:
Do not release untreated into natural waters.

The product has not been tested. The statements on ecotoxicology have been derived from the properties of the individual components.

13. Disposal Considerations

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:
Uncontaminated packaging can be re-used.

Packs that cannot be cleaned should be disposed of in the same manner as the contents.

14. Transport Information

Domestic transport:

UN number or ID number	Not classified as a dangerous good under transport regulations
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable

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Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
Proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

15. Regulatory Information

Other regulations

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

16. Other Information

Information on intended use: This product is of industrial quality and unless otherwise specified or agreed intended exclusively for industrial use. This includes the mentioned and recommended usage. Any other intended applications should be discussed with the manufacturer. In particular this concerns the application for products that are the object of special standards and regulations.

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Vertical lines in the left hand margin indicate an amendment from the previous version.

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